

Talking with Neurons

Prasanth Kunasilan
prasanth.kk@icloud.com
National University of Singapore
Singapore

Alexandra Arguelles
corrine.arguelles@gmail.com
National University of Singapore
Singapore

Yuri Klebanov
yurikleb@iis.u-tokyo.ac.jp
Institute of Industrial Science
The University of Tokyo
Tokyo, Japan

Siu Yu Angela Chow
angelach@iis.u-tokyo.ac.jp
Institute of Industrial Science
The University of Tokyo
Tokyo, Japan

Huaruo Hu
huaruo@iis.u-tokyo.ac.jp
Institute of Industrial Science
The University of Tokyo
Tokyo, Japan

Tomoya Duenki
duenkit@iis.u-tokyo.ac.jp
Institute of Industrial Science
The University of Tokyo
Tokyo, Japan

Atsuhiko Nabeta
atsuhiko-nabeta@g.ecc.u-tokyo.ac.jp
Institute of Industrial Science
The University of Tokyo
Tokyo, Japan

Yoshiho Ikeuchi
yikeuchi@iis.u-tokyo.ac.jp
Institute of Industrial Science
The University of Tokyo
Tokyo, Japan

Miles Pennington
miles@iis.u-tokyo.ac.jp
Institute of Industrial Science
The University of Tokyo
Tokyo, Japan

Hyunjung Kim
hyunjung@iis.u-tokyo.ac.jp
Institute of Industrial Science
The University of Tokyo
Tokyo, Japan

ACM Reference Format:

Prasanth Kunasilan, Alexandra Arguelles, Yuri Klebanov, Siu Yu Angela Chow, Huaruo Hu, Tomoya Duenki, Atsuhiko Nabeta, Yoshiho Ikeuchi, Miles Pennington, and Hyunjung Kim. 2023. Talking with Neurons. In *Proceedings of SIGGRAPH '23 Art Gallery*. ACM, New York, NY, USA, 3 pages. <https://doi.org/10.1145/3588428.3593825>

1 PROJECT URL

<https://www.designlab.ac/neurons>

2 DESCRIPTION OF ARTWORK

"Talking with Neurons" is an interactive installation that enables people to remotely connect with live neural tissues that mimic our nervous system. Audiences activate lab-grown neural circuits called neural organoids in an off-site lab through their vocal cues. The neurons respond to this stimulation by exhibiting patterns of activity that are subsequently translated into audiovisual representations. Essentially, people talk to the neurons, which then promptly respond to the audience.

The configuration of "Talking with Neurons" consists of an installation virtually connected between an on-site exhibition and a remote lab in Tokyo. The on-site location is equipped with a

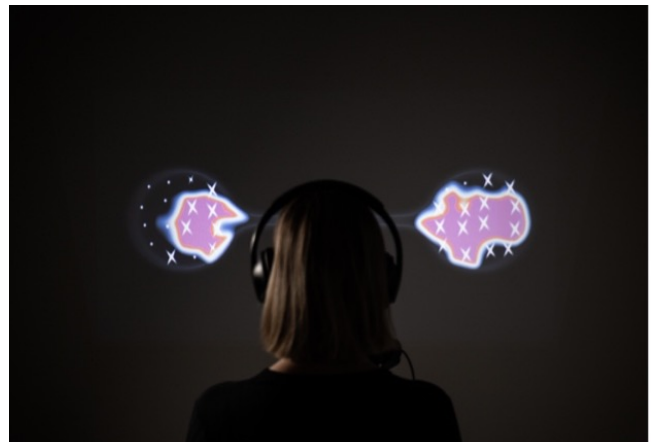
Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

SIGGRAPH '23 Art Gallery, August 06-10, 2023, Los Angeles, CA, USA

© 2023 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-0157-3/23/08.

<https://doi.org/10.1145/3588428.3593825>



projector, screen, and headset to allow the audience to digitally interact with the neurons. In the virtual setup, in vitro neural circuits are connected to a microelectrode array (MEA) that stimulates the organoids based on the remote input – the audience's voice – and records neuronal activity patterns in response to the stimulation. The captured signals are then transmitted to the on-site computer in real-time, and instantly converted into auditory and visual data.

The aural interaction between the user and in vitro neurons is based on intonation, as changes in pitch and rhythm allow individuals to express meaning beyond words. Key activity markers of the neuronal patterns, such as isolated spikes representing rapid and transient electric signals generated by neuron excitation, and bursts



which consist of a group of spikes within a small time duration, are encoded into simple sounds. As a result, users can experience the inception of basic communication with lab-grown organoids.

Audiences are given the opportunity to witness a stunning visual rendition of neuronal harmony, demonstrating the spatial and temporal intricacies of the neuronal activities exhibited by in vitro neural circuits. The attached neural organoids, depicted in an abstract manner as two large circles bridged with a thin line surrounded by a dark background, contrast with bright flares that light up during burst events and glimmers that pulsate in time with neuronal spikes. The abstracted visuals of the connected organoid's activity are juxtaposed with live footage of the experiment conducted at the remote lab. The installation re-imagines the technical, data-based experiments as an expressive and engaging experience by relaying complex knowledge to the audience.

With each trial, our systems summarize the visual responses through snapshots of the interaction, adding them to an online archive of these compelling conversations between humans and living neurons. What will be revealed as the neural tissues continue to engage in further conversations? Will they respond consistently to similar auditory stimuli? Will they learn and develop? We will ensure that the archive remains accessible to everyone online and contributes to our understanding of neurons, which are fundamental to our nervous system and, consequently, our minds.

"Talking with Neurons" is the result of a collaboration between neuroscientists and artists that enables the general public to engage with in vitro neural circuits. By granting users access to cutting-edge neuroscience research that is typically inaccessible to the public, we hope that "Talking with Neurons" will foster lively discussions about the future of biocomputing, as well as its artistic possibilities, where living neurons are integrated into the technology we use every day.

3 ARTIST INFORMATION

Prasanth Kunasilan is an experienced designer who believes that good design should be accessible to all. He loves to explore various disciplines, such as motion design and interaction design, in order to create a more holistic user experience. He enjoys tinkering with electronics and new digital media to bring his designs

to life. Prasanth will be graduating from the National University of Singapore with a Bachelor of Arts in Industrial Design in 2023. Arguelles believes that design is a way to explore the world around us. With a particular interest in creative direction and experience design, she hopes to make a positive impact through authentic and joyful interactions. Alexandra will be graduating from the National University of Singapore with a Bachelor of Arts in Industrial Design in 2023. Yuri Klebanov graduated from the Department of Visual Communication at Bezalel in 2011. While working as an Interaction Designer at Innovid, he built interactive installations as a hobby, which were showcased at festivals around Israel and Europe. In 2016, Yuri received his MA and MSc in Innovation Design Engineering from the Royal College of Art and Imperial College London, respectively. He then worked as a creative developer at Hirsch&Mvccwwe Socila Msoci in London. Currently, Yuri is a design and technology researcher at the Institute of Industrial Science Design Lab at the University of Tokyo, where he focuses on merging design and science. Siu Yu Angela Chow joined Ikeuchi Lab in 2017, where she earned her Ph.D., after graduating from McGill University in Canada. She is currently working as a Project Research Associate in Ikeuchi Lab after serving as a Project Researcher for one year. Her current research interest lies in investigating the brain's reward system by connecting different brain regions based on the lab's tissue model. Huaruo Hu is a master student at Ikeuchi Lab who has completed her bachelor's degree at Northeastern University and is currently working on a project to explore the interface of neurobiology and artificial intelligence. Tomoya Duenki is a Ph.D. student at the Ikeuchi Lab who aims to leverage advances in biomedical sciences to create new technologies for a better understanding of human brain function. He received his master's degree in Biomedical Engineering from ETH Zurich in Switzerland and has gained hands-on research experience in neuro-engineering and biomanufacturing at the RIKEN Center for Brain Science and the Wyss Institute for Biologically Inspired Engineering at Harvard University. Atsushi Nabeta has worked as a system architect in various system developments, including web applications and smart device applications. In this project, he implemented a server that transfers voltage data from neurons to the presentation, as well as a server that feeds back audio data from the presentation to neurons. Yoshiho Ikeuchi earned his Ph.D. degree from The University of Tokyo. He completed post-doctoral neuroscience training at Harvard Medical School and Washington University in St. Louis. In 2014, he was appointed as a Lecturer, and later promoted to Associate Professor in 2018 at the Institute of Industrial Science, The University of Tokyo. His research goal is to create nervous tissue (organoids) from human multipotent stem cells (iPS cells) and to make them functional in order to better understand the mechanisms of brain circuitry. Miles Pennington is a Professor of Design Led Innovation at The University of Tokyo. He is also the Director of the DLX Design Lab, which collaborates with science and technology researchers to create value through design. Previously, he worked at the Royal College of Art, where he served as the Head of the Program of the Innovation Design Engineering joint Masters program with Imperial College. He is an alumnus of the IDE program and graduated in 1992. Additionally, he was the Founder and Head of the international exchange program Global Innovation Design program. Before joining The

University of Tokyo, he was a Director of the London office of the innovation consultancy Takram. Currently, he also serves as a Specially Appointed Advisor to the UTokyo President, focusing on promoting GX International Cooperation and recurrent education strategy. Hyunjung Kim is a designer, researcher, and educator who designs and experiments with interactive products that explore the future of everyday life. Her work has garnered attention at international conferences such as ACM SIGGRAPH and CHI, and she has received several design awards, including the Red Dot: Best of the Best and the iF Concept Design Award. She majored in Industrial Design at the Korea Advanced Institute of Science and Technology (KAIST), where she also received her Ph.D. in 2013. She has

over a decade of wide-ranging professional experience across several leading research institutions, including NASA Ames Research Center and Sony Computer Science Laboratories. Currently, she is a project researcher at the DLX Design Lab at the Institute of Industrial Science at The University of Tokyo, where she conducts innovative science and design collaboration projects.

ACKNOWLEDGMENTS

This work was supported in part by the UTokyo Ushioda Foundation, Institute for AI and Beyond, and the Foundation for the Promotion of Industrial Science.